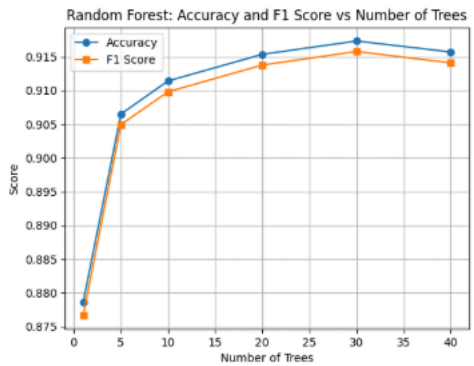
The Rice Grains Dataset

Random Forest (Adam)

Hyperparameter testing:

ntree	Accuracy	F1
1	0.8786	0.8766
5	0.9065	0.9049
10	0.9114	0.9098
20	0.9154	0.9138
30	**0.9173**	**0.9158**
40	0.9157	0.9141



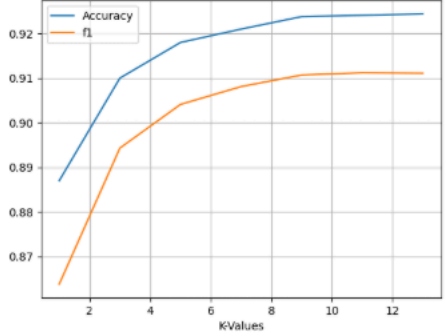
On the rice grains dataset, the performance steadily improves as the number of trees increases, peaking at around ntree = 30. The accuracy and F1 scores plateau and even decrease slightly, suggesting diminishing returns from adding more trees. This makes sense given the dataset contains only numerical features, which Random Forests handle very well without extra preprocessing. The model likely captures relevant splits early, and additional trees just refine rather than transform its performance. There's no sign of overfitting here, so Random Forest is a good match for this dataset.

The Rice Grains Dataset

KNN (Sahil)

Hyperparameter testing: [1,3,5,7,9,11,13]

K Value	Accuracy	F1
1	.887	.8638
3	.910	.8943
5	.918	.9041
7	.921	.9081
9	.9238	.9107
11	.9241	.911
13	**.9244**	**.9111**

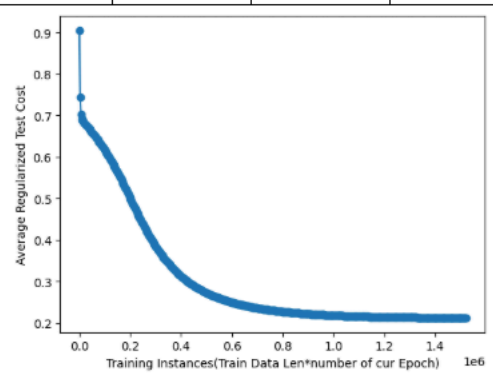


KNN performs well on the dataset where it reaches over 92.44% accuracy and 91.11% F1 score at k = 13. Since the data only contains numerical features it is great for the distance based method of KNN. The performance improves steadily as K increases and plateaus after k = 7. This trend suggests that including more neighbors would just stabilize performance as it is seen to plateau, and would serve little purpose other than to increase runtime.

The Rice Grains Dataset

Neural Networks (Sahil)

Learning Rate	Regularization	Network Architecture	Accuracy	F1	Epoch
1	0.1	7,1	.922	.907	300
1	0.1	7,3,1	.925	.912	300
1	0.3	7,3,1	.927	.9145	300
1	0.1	7,5,3,1	.9257	.913	300
1	**0.1**	**7,5,1**	**.928**	**.917**	**500**
1	0.1	7,3,1	.928	.916	500



Neural networks perform very well on the Rice dataset. Results improve slightly with deeper networks and longer training, particularly for [7,5,1] and [7,3,1]. The dataset's fully numerical features allow the network to learn effectively without having to deal with one hot encoding. The training curve shows a steep decline in regularized test cost that gradually plateaus, indicating that the model benefits from more training data but eventually reaches a plateau. This suggests the model is converging well without overfitting and with more training instances, improves the model. The graph was constructed using the [7,5,1] architecture with 500 epochs, and the plateau seen suggests that lower epochs could be used if we want to minimize runtime and compute spent. The fully numerical data that Rice has makes NNs the perfect algorithm to run on it.